



OPEN Clinical-guided deep learning framework for diabetic retinopathy: integrating lesion-aware attention, adversarial augmentation, and uncertainty quantification

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The global prevalence of diabetic retinopathy (DR) is increasing in parallel with the rising burden of diabetes, posing a substantial public health challenge. Current automated DR screening systems often lack lesion-level interpretable, robustness against class imbalance, and reliable uncertainty estimation, limiting their applicability in real-world clinical settings. The study suggests a clinical-guided deep learning framework of diabetic retinopathy (CG-DRNet) that considers lesion-conscious attention, adversarial data augmentation, and Bayesian uncertainty measurement to achieve reliable and explainable DR severity detection with objective to allow early-stage detection, particularly mild nonproliferative diabetic retinopathy (NPDR), and reciprocate clinical diagnostic procedures. The proposed framework uses a multi-task deep learning framework and lesion-aware attention network to explicitly predict microaneurysms, hemorrhages, exudates, and neovascularization. The system consists of a conditional generative adversarial network (CWGAN-GP) that is employed to overcome extreme imbalance in classes by generating clinically realistic images of minority classes of the fundus. Monte Carlo dropout is used to model Bayesian uncertainty to estimate predictive confidence, and an uncertainty-informed semi-supervised strategy of learning is used to enhance data efficiency. Evaluation of the framework is carried out on publicly available fundus image datasets APTOS 2019, Messidor-2 and Clinical metadata with the use of PyTorch 2.0.1. The proposed CG-DRNet reached 93.8% accuracy on APTOS 2019 and 91.2% on Messidor-2 with only 2.6% difference between generalization and the original model, macro F1-score of 0.891, quadratic weighted kappa of 0.912, referable DR detection AUC of 0.963 and expected calibration error of 0.034. The framework was shown to have 84.7% sensitivity when detecting Grade 2+ with 67 ms inference time and therefore has been shown to be viable in clinical use.

Keywords Diabetic retinopathy, Fundus imaging, APTOS 2019, Uncertainty quantification, Monte Carlo dropout

It is projected that by 2040, close to 600 million people will be living with diabetes with one-third of them likely to develop diabetic retinopathy (DR) as the most common cause of vision loss among working-age adults worldwide¹. DR is a neurovascular complication in which processes of AGE accumulation, PKC activation and VEGF upregulation participate thus degrading the inner blood-retinal barrier². Typical appearances are microaneurysms, intraretinal hemorrhages, hard and soft exudates and neovascular proliferation. Some forms of DR is presently experienced by approximately 90 million people and over 30 million have vision threatening conditions including proliferative DR (PDR) and diabetic macular edema (DME)³. The international diabetes federation estimates that diabetic patients will surpass 700 million by 2045 which will carry an unparalleled burden on healthcare systems especially in resource constrained environments⁴.

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There is a lot of clinical importance in early diagnosis. The landmark DCCT and UKPDS trials revealed that HbA1c, when kept below 7%, decreases the risk of DR by up to 76% in type 1 diabetes and 25% in type 2 diabetes. Regardless of this, the global coverage of screening has been low because of labour shortage, geographic differences and other inter-observer variability, which highlights the necessity to develop scalable automated screening solutions⁵. DR is analyzed according to international clinical diabetic retinopathy severity scale, which is based on Mild NPDR (microaneurysms only) and Moderate and Severe NPDR defined by the ETDRS 4-2-1 rule to PDR, where the neovascularization is necessitated by urgent treatment⁶. Manual grading is subjective, time-consuming, and inaccurate, with the consensus of the experts being as low as 70–80% of cases and throughput mostly restricted to 100–120 patients daily which is a long way short of the needs of large-scale screening⁷. The solution is provided by deep learning, especially Convolutional Neural Networks (CNNs). CNNs can be trained on large datasets with labeled labels that have proven to perform at the level of an expert in medical imaging, which has strongly suggested the use of these in automated screening and grading of DR⁸. In the world, diabetic related retinopathy is the main cause of blindness particularly in countries that are resource deprived. This is the reason why a good and precise diagnostic system is highly required. Figure 1 demonstrates different phases of diabetic retinopathy.

Current DR systems have a number of limitations. First, the majority of those focus on the crude binary classification, where mild NPDR and no DR fall into non-referable categories, missing an opportunity of the critical early intervention, proven by the ACCORD and DCCT/EDIC trials⁹. Second, lesion-level reasoning is absent in end-to-end black-box classifiers, thus decreasing interpretability and clinician trust, which is an important obstacle to clinical adoption. Third, standard models do not quantify uncertainty, which leads to overconfident prediction over uncertain cases; Bayesian methods such as Monte Carlo dropout can do the same but are mostly not used in DR systems¹⁰. Fourth, a small amount of expertly annotated datasets create the risk of overfitting and the lack of sensitivity to infrequent lesions. Fifth, conventional active learning has logistical and cost limitations in a pipeline that exist in reality. Sixth, most of the models do not consider co-occurrence and patterns of progression of a structured lesion, thus, throwing away good contexts. Seventh, extreme imbalance in classes worsens performance of key minority classes such as PDR. Eighth, models show severe performance impairment in domain shift in a variety of clinical settings.

It has been proposed to overcome these complex and integrated challenges that Clinical-Guided Deep Learning Framework to DR. Integrating lesion-aware attention, adversarial augmentation, and uncertainty quantification mainly known as clinical-guided diabetic retinopathy network (CG-DRNet) has been proposed that integrates four synergistic technical innovations: (1) a lesion-aware attention mechanism that models multi-lesion segmentation as an auxiliary task to generate interpretable diagnostic features (2) an adversarial augmentation strategy employing conditional generative adversarial network. (3) Bayesian uncertainty quantification via Monte Carlo dropout, and (4) an uncertainty-guided active learning approach that leverages

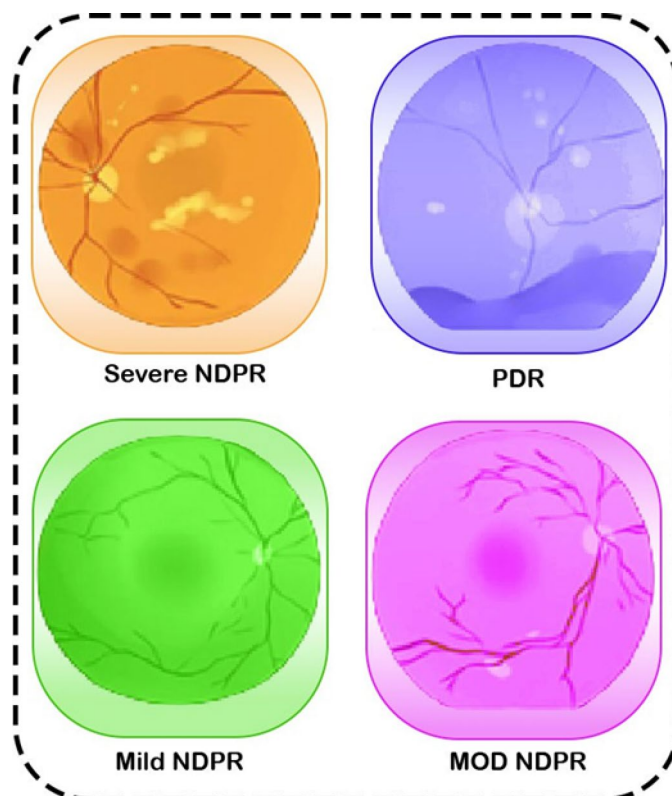


Fig. 1. Different levels of diabetic retinopathy.

predictive uncertainty to automatically select informative unlabeled samples for pseudo-labeling without requiring human annotation.

A multi-task learning model explicitly represents lesion detection and segmentation, which explicitly captures hierarchical diagnostic structure and has interpretable representations, excluding spurious correlations. The suggested (CG-DRNet) framework contributes critically important key contributions which include:

- i. Having a multi-lesion segmentation pathway that can be an auxiliary task of the main DR classification task, which integrates new attention mechanisms that explicitly encode spatial and semantic relationships between various lesion types (microaneurysms, hemorrhages, hard exudates, soft exudates) and maintains fine-scale spatial information that is essential in the accurate detection of small lesions such as microaneurysms that can be only 15–30 pixels in diameter.
- ii. Proposed an adversarial augmentation model, named as conditional Wasserstein GAN with gradient penalty (CWGAN-GP) that learns to synthesize realistic fundus images under the condition of target DR disease severity category and lesion desired traits, and the specifics of which are to augment minority classes (severe NPDR and PDR), but without compromising the clinically-realistic lesion morphology, spatial distribution, and co-occurrence with other lesions.
- iii. Bayesian uncertainty quantification Implementation in the form of Monte Carlo dropout on the lesion segmentation pipeline and the DR classification pipeline, where several stochastic forward passes are performed at inference time to draw samples out of the approximate posterior distribution over predictions and calculate predictive entropy.
- iv. Designing an uncertainty-guided active learning strategy that leverages the computed uncertainty scores to automatically select the most informative unlabeled samples from a large pool, generates pseudo-labels for these samples using the current model's predictions when uncertainty falls below a specified threshold.

Rest of the sections are arranged as: The comprehensive research gap in the previous studies has been explored in Sect. “[Literature review](#)”. Model design and clinical data integration is discussed in Sect. “[Clinical data integration workflow](#)”. Section “[Experimental setup](#)” details the experimental setup and configurations. Result and performance comparison with other models is discussed in Sect. “[Experimental results with performance analysis and discussion](#)”. Findings are concluded in the “[Conclusion](#)” section with future research directions.

Literature review

An overview of the latest research on deep learning reveals that automated diagnosis of DRs has shortcomings, such as lack of robustness, uncertainty estimation and integration of clinical guidance. The problem of medical image classification of diabetic retinopathy is a great challenge because masses of irrelevant data are included in fundus images which need attention mechanism in order to focus on the important areas of diagnosis. Both the necessity of correct classification and interpretable visualizations that are consistent with clinical reasoning were essential. Wang et al.¹¹, proposed a multi-attention residual refinement architecture with three strategic modifications. The framework incorporated class-specific multi-attention with temperature-weighted spatial features, space-to-depth preprocessing for lossless resolution conversion, and Squeeze-and-Excitation blocks for channel-wise optimization, validated across ResNet50, DenseNet121, EfficientNet-B0, and MobileNetV3 using EyePACS with transfer learning. ResNet50 achieved sensitivity of 0.80, specificity of 0.81, Kappa of 0.82, and AUC of 0.89, with Squeeze-and-Excitation blocks contributing most in ablation studies. However, validation was limited to EyePACS with minimal external data, uncertainty quantification was absent, and proliferative DR classification remained poor, with AUC of only 0.64 and frequent misclassification as moderate or severe DR. Major shortcomings in this study are prospective clinical validation, lesion segmentation integration, and comprehensive computational analysis including inference time and memory were observed.

Explicit modeling of retinal lesions is increasingly recognized as essential for clinically interpretable diabetic retinopathy systems. Within this context, Karuppusamy et al.¹², propose a lesion-aware hybrid attention framework (DL-LAHA) that combines segmentation and classification in a unified pipeline. The architecture utilizes recursive sub-image histogram equalization to enhance the images, Nested V-Net to obtain lesion segmentation and dual-backbone feature extraction by incorporating EfficientNet and RegNetY. The feature selection is done using entropy which is followed by classification so that specific information related to lesions is highlighted. The model attains a high accuracy of 98.65 per cent on the MESSIDOR dataset, which is much higher than a number of traditional CNN-based methods. The approach is able to explain the pathological relevance and interpretability of predictions by explicitly applying lesion segmentation, which is ideally applicable to explainable clinical screening. The complexity of the pipeline is however a weakness since it propagates errors and fails to include uncertainty estimation, the biggest deficiency of the study.

The problem of decreasing the cost of annotation but preserving the quality of diabetic retinopathy classification is a continuous dilemma in medical imaging. To deal with this problem, Smailagic et al.¹³, introduce an active learning system, combining Bayesian uncertainty estimation with deep CNN to classify DR. The model itself is an iterative selection of the most informative unlabeled samples according to predictive entropy and verified by the expertise and included in the training set. The model can also learn more efficiently with fewer labeled images and still achieve competitive performance with this strategy. Empirical evidence indicates that the method performs state-of-the-art classification performance with a very high level of reduction in annotation needs indicating the validity of uncertainty-based selection of samples. The paper highlights the significance of uncertainty in predictability as well as effective dataset development in large scale screening systems. The primary weakness is the lifelong expertise and intricate training pipelines which will ultimately restrain the size.

In the context of improving the clinical reliability and interpretability of deep learning-based DR diagnosis, Ling Dai et al.¹⁴, introduce an attention-guided framework, which improves the discriminative features learning

process, specifically, by targeting retinal areas related to lesions. Their work is inspired by the fact that traditional convolutional neural networks tend to use global image representations and this may sometimes blur fine pathological features that are important in the proper grading of DR. The model has been suggested, and it provides the attention mechanism in a deep convolutional architecture, which allows the network to selectively highlight informative retinal objects and deactivate useless background areas. The framework is tested on publicly available fundus image datasets, where it can be seen to be more accurate and robust in classification than the baseline CNN models. The quantitative findings demonstrate the higher sensitivity and specificity of the research at various levels of severity of DR, meaning the ability to recognize finer-grained pathological patterns. Besides, attention visualizations also offer intuitive explanations which can be seamlessly interpreted ophthalmologically, which explains clinical interpretability. The paper helps to fill the gap between automated DR screening systems and clinical needs because it focuses on lesion-based learning and predictability. The framework fails to quantify uncertainty hence undermining its ability to be robust and reliable in the presence of a shift in distribution.

Diagnosis of DR is not easily automated as the clinician must effectively record the global retinal structures as well as the detailed lesion-level information at different stages of the disease. Proposed by Feng Li et al.¹⁵, is a deep learning architecture that improves the feature representation by and with the use of multi-scale learning and attention-oriented mechanisms on retinal fundus imagery. A multi-branch architecture by authors is based on hierarchical characteristics at different spatial resolutions that integrates attention modules within the network and dynamically highlights pathological conditions like microaneurysms and hemorrhages. They compared EyePACS and Messidor, the model performs better than the traditional CNNs on the basis of accuracy, sensitivity, and specificity, especially on moderate and severe DR. Visualizations These can ensure that the images are focused on the abnormalities found by the ophthalmologist, enhancing clinical interpretability and intensifying features of attention-based, multi-scale learning on retina screening. Nevertheless, method does not quantify uncertainty and has adversarial robustness, and this can have an impact on reliability when there are changes in the real-world distribution.

A correct DR grading demands proper modeling of the global retinal structures and local pathological areas, and this is still difficult using conventional convolutional neural networks. The study proposed by Lei-Si Hu et al.¹⁶, is a DR classification architecture implemented using deep learning and includes attention mechanisms to improve the ability to learn discriminative features based on fundus images. The attention-guided network is an adaptive focus on the lesion-relevant features, which rejects the background noise to enhance the data representation of subtle abnormalities such as microaneurysms and hemorrhages. The model has been evaluated on publicly available datasets of DR, and it is more accurate and sensitive than the standard CNN baselines when evaluated on multiple severity grades of lesions, especially those that are sparsely distributed or ambiguous. Image-focused attention maps show that they are well aligned with ophthalmologist defined clinically significant areas and these maps are interpretable and add to explainable deep learning solutions to DR screening. Yet uncertainty quantification like adversarial augmentation is not part of this framework, which makes it less reliable in the context of clinical variability in reality.

The reviewed studies highlight the important advances in lesion-aware learning, attention mechanisms, and uncertainty-based diabetic retinopathy analysis; however, most approaches address these challenges separately and do not provide a unified solution for uncertainty awareness, interpretability, robustness, and clinical applicability. To address all these limitations, the proposed CG-DRNet integrates lesion-aware attention for clinically meaningful feature localization, adversarial augmentation for class imbalance mitigation, Bayesian uncertainty quantification for confidence-aware decision support, and clinical meta-data fusion for personalized risk assessment. Table 1 summarizes how the proposed framework directly addresses the major limitations identified in previous studies.

Studies	Method	Dataset used	Limitations	Proposed solution
Want et al.	Multi-attention residual refinement	EyePACS + 1800 clinical images	Poor performance on proliferative DR	Lesion-aware attention combined with adversarial augmentation to improve sensitivity for underrepresented advanced DR stages and enhance minority-class learning
Karuppusamy et al.	Lesion-aware hybrid attention framework (DL-LAHA)	MESSIDOR	High model complexity, lack of uncertainty estimation, potential overfitting	End-to-end lesion-aware architecture integrated with Bayesian Monte Carlo Dropout for uncertainty quantification and improved clinical reliability
Smailagic et al.	Active learning with Bayesian uncertainty estimation	Not specified	limited scalability	Uncertainty-guided semi-supervised learning that leverages predictive confidence for pseudo-label generation, reducing dependence on repeated expert annotation*
Ling Dai et al.	Attention-guided deep convolutional framework	Public fundus datasets	No uncertainty modeling, limited real world robustness	Integration of lesion-aware attention, Bayesian uncertainty estimation, and adversarial augmentation to improve interpretability, confidence estimation, and robustness
Feng Li et al.	Multi-scale learning with attention guided mechanisms	EyePACS, Messidor	No uncertainty quantification, limited real-world robustness	Combination of lesion-aware attention, adversarial augmentation, and external validation to improve robustness, generalization, and performance under do-main shift
Lei-Si Hu et al.	Attention-guided deep learning network	Public DR datasets	No uncertainty modeling, lacks robustness strategies	Clinical-guided framework incorporating Monte Carlo Dropout-based uncertainty quantification, adversarial augmentation, and clinical metadata fusion for reliable deployment in real-world screening settings

Table 1. Summary of reviewed diabetic retinopathy classification methods, identified limitations, and proposed solution.

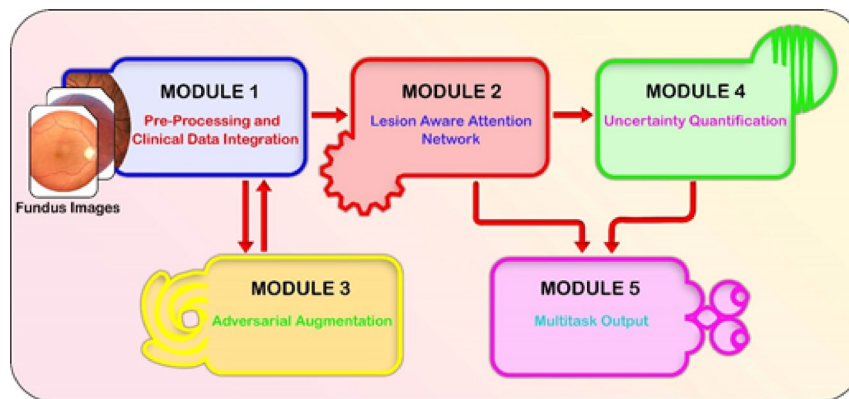


Fig. 2. The overview of the architecture of the CG-DRNet, which features image pre-processing, lesion-aware feature extraction, uncertainty estimation, clinical metadata fusion, and diabetic retinopathy classification.



Fig. 3. Lesion-aware dual attention network architecture.

Proposed CG-DRNet framework

The proposed CG-DRNet framework has a comprehensive architecture and integration methodology which is discussed in Fig. 2. The architecture comprises of five modules that are interconnected and collaborate with each other to provide sound, interpretable, and clinically sound DR grading.

The proposed CG-DRNet architecture is shown in Fig. 2. In Module 1, fundus images and related clinical data are preprocessed and compiled. The processed inputs are then fed into a lesion-aware attention network (Module 2) that emphasizes retinal lesions associated with the disease for discriminative feature learning. Adversarial augmentation (Module 3) increases data diversity and model robustness, while the uncertainty quantification module (Module 4) quantifies prediction confidence. The results of these modules are then integrated at the last module, called the multitask learning module (Module 5) in order to obtain reliable results for the detection and grading of DR.

The system combines four main elements: (1) lesion-aware dual attention network of feature extraction, (2) adversarial augmentation pipeline of robust data generation, (3) Bayesian uncertainty quantification of confidence estimation and (4) clinical metadata fusion of personalized risk assessment. All the components are intended to mitigate certain issues of automated DR screening as well as ensure clinical interpretability and compliance to regulations.

Lesion-aware dual attention network

The architecture is the main characteristic extraction mechanism of CG-DRNet framework, having a clear design of detecting both fine-grained pathological and broad-scale retinal structures. The proposed solution compared to the rest of the convolutional architectures that assume that all locations in space are treated equally, introduces complementary attention mechanisms on various semantic levels.

The network architecture comprises three sequential stages depicted in Fig. 3: (1) backbone feature extraction using pre-trained EfficientNet-B4 (2) spatial attention module emphasizing lesion-containing regions, and (3) channel attention module amplifying pathology-discriminative feature channels. The dual attention mechanism

enables the network to answer where to look (spatial) and what to emphasize (channel) when processing fundus images with variable lesion distribution and severity.

The EfficientNet-B4 has been adopted as the feature extraction backbone due to its optimal accuracy-efficiency trade-off achieved through compound scaling of network depth, width, and resolution. The architecture processes input fundus images $I \in \mathbb{R}^1$, where $H = 380$ and $W = 380$ pixels, after preprocessing through a sequence of mobile inverted bottleneck convolution (MBConv) blocks with squeeze-and-excitation optimization. Pre-trained weights from ImageNet are transferred and fine-tuned on fundus images, enabling the model to leverage learned representations of edges, textures, and spatial hierarchies that are relevant to biological imaging.

The feature extraction process is formulated as Eq. (1):

$$F = \varphi_{\text{backbone}}(I; \theta_E) \quad (1)$$

where $F \in \mathbb{R}^{(h \times w \times c)}$ represents the extracted feature maps, i.e. $h = 12$ and $w = 12$, spatial dimensions, $c = 1792$ the number of output feature channels generated by EfficientNet-B4. Here, θ_E denotes the backbone parameters, and $\varphi_{\text{backbone}}$ encapsulates the sequential transformations through MBConv blocks.

Spatial attention module

The spatial attention module generates location-specific attention weights that highlight regions containing diabetic retinopathy (DR)-related pathological features. It diagnoses lesions such as microaneurysms, hemorrhages, exudates, and neovascularization within fundus images.

Feature maps F from the backbone is applied to global context pooling to create descriptor vectors summarizing information across channels in Eqs. (2) and (3)

$$F_{\text{avg}} = \text{GlobalAvgPool}(F) \in \mathbb{R}^{h \times w \times 1} \quad (2)$$

$$F_{\text{max}} = \text{GlobalMaxPool}(F) \in \mathbb{R}^{h \times w \times 1} \quad (3)$$

Both average and max pooling operations compress channel information while preserving spatial structure. The concatenated descriptors $[F_{\text{avg}}; F_{\text{max}}]$ are fed through a convolutional layer with sigmoid activation to produce spatial attention map given in Eq. 4

$$M_{\text{spatial}} = \sigma(\text{Conv}_{7 \times 7}([F_{\text{avg}}; F_{\text{max}}]; \theta_s)) \quad (4)$$

where, $M_{\text{spatial}} \in \mathbb{R}^{h \times w \times 1}$ contains spatial attention weights in range $[0, 1]$, θ_s represents learnable convolutional parameters, $\text{Conv}_{7 \times 7}$ denotes 7×7 convolution for large receptive field, and σ is the sigmoid function ensuring weights are normalized.

The spatial attention mechanism learns to assign high weights to regions containing DR lesions through supervised training.

During training, the network observed feature activations, corresponding to microaneurysms, hemorrhages, and exudates correlate with severity labels, thereby upweighting those locations in M_{spatial} . This creates an implicit lesion localization capability.

The spatially-refined features are computed by element-wise multiplication in Eq. 5.

$$F^{\text{spatial}} = F \odot M_{\text{spatial}} \quad (5)$$

where $\odot$ denotes element-wise multiplication with broadcasting. This operation amplifies feature activations in diagnostically important regions.

Channel attention module

While spatial attention determines where to attend, channel attention addresses what features to emphasize by recalibrating channel-wise feature responses. In fundus image analysis, different convolutional channels activate for different visual patterns; some channels respond to vascular structures, others to exudates (bright yellow lipid deposits), hemorrhages (dark red blood), or microaneurysms (small focal lesions).

Given the spatially refined features F^{spatial} , squeeze-and-excitation (SE) operations are applied to model inter-channel dependencies.

Squeeze phase

Global spatial information is compressed into channel-wise descriptors using global average pooling, as defined in Eq. (6):

$$z = \frac{1}{h \times w} \sum_{i=1}^h \sum_{j=1}^w F^{\text{spatial}}(i, j) \quad (6)$$

where $z \in \mathbb{R}^c$ represents the channel descriptor vector that summarizes global statistical information for each of the c feature channels.

Excitation phase

The channel descriptor is transformed through a two-layer feedforward network to generate channel attention weights in Eq. (7):

$$M_{channel} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)) \tag{7}$$

where $W_1 \in \mathbb{R}^{\bar{r}c \times c}$ and $W_2 \in \mathbb{R}^{\bar{r}c \times c}$ are learnable weight matrices, $r=16$ is the reduction ratio controlling the bottleneck dimension, ReLU introduces non-linearity, and σ sigmoid activation producing weights in $[0, 1]$ range.

The channel-recalibrated features are computed in Eq. (8):

$$F_{channel} = F_{spatial} \odot M_{channel} \tag{8}$$

where the broadcasting ensures $M_{channel} \in \mathbb{R}^c$ is multiplied across all spacial locations, scaling each channel uniformly across the feature map.

Attention fusion and classification

The final attended features, $F_{channel}$, combine both spatial and channel attention, creating representations that are simultaneously localized. These refined features feed into the classification head for DR severity grading and DME detection:

Classification architecture

The classification module consists of the following components:

- Global average pooling: $F_{channel} \rightarrow \mathbb{R}^c$, reduces spatial dimensions to fixed-length vector
- Fully connected layer 1: $\mathbb{R}^c \rightarrow \mathbb{R}^{512}$ with ReLU activation and Dropout ($p=0.5$).
- Fully connected layer 2: $\mathbb{R}^{512} \rightarrow \mathbb{R}^{256}$ with ReLU activation and Dropout ($p=0.3$).
- Output layer (multi-task):
 - DR severity head: ($\mathbb{R}^{256} \rightarrow \mathbb{R}^5$) with softmax for 5-class classification (Grades 0–4), DME Detection head ($\mathbb{R}^{256} \rightarrow \mathbb{R}^1$) with sigmoid for binary classification.
 - DME detection head: $\mathbb{R}^{256} \rightarrow \mathbb{R}^1$ using sigmoid activation for binary classification.

The multi-task learning formulation enables the joint optimization of DR severity grading and DME detection¹⁷, sharing the attention-enhanced representations while maintaining task-specific output heads. The combined loss function is given in Eq. (9):

$$L_{total} = L_{DR} + \lambda \cdot L_{DME} \tag{9}$$

where L_{DR} is categorical cross-entropy loss for severity grading, L_{DME} is binary cross-entropy for DME detection, and $\lambda=0.5$ balances task importance. Multi-task learning enhances generalization because it makes the network to learn representations that are useful in both related and different diagnostic tasks. Table 2, gives the architectural specifications of Lesion-Aware Dual Attention Network.

Component	Layer type	Input dimension	Output dimension	Parameters	Activation
Backbone	EfficientNet-B4	380×380×3	12×12×1792	17.7 M	Swish
	MBCConv Blocks	12×12×1792	12×12×1792	16.2 M	Swish
	Global Avg Pool	12×12×1792	1792	0	–
Spatial attention	Global Avg Pool	12×12×1792	12×12×1	0	–
	Global Max Pool	12×12×1792	12×12×1	0	–
	Conv 7×7	12×12×2	12×12×1	98	Sigmoid
	Element-wise $\odot$	12×12×1792	12×12×1792	0	–
Channel attention	Global Avg Pool	12×12×1792	1792	0	–
	FC: 1792→112	1792	112	200 K	ReLU
	FC: 112→1792	112	1792	202 K	Sigmoid
	Element-wise $\odot$	12×12×1792	12×12×1792	0	–
Classification head	FC: 1792→512	1792	512	917 K	ReLU + Dropout(0.5) ReLU + Dropout(0.3)
	FC: 512→256	512	256	131 K	
	Total Trainable	–	–	~ 19.6 M	–
Output heads	DR Severity: 256→5	256	5	1,285	Softmax
	DME Detection: 256→1	256	1	257	Sigmoid

Table 2. Lesion-aware dual attention network specifications.

Adversarial augmentation pipeline

The DR datasets also have severe data paucity problems especially in severe NPDR (Grade 3)¹⁸, proliferative DR (Grade 4) which is a clinically relevant population, but is not overrepresented in screening populations. Moreover, fundus photographic images exhibit high inter-dataset variability because they may vary in terms of camera used, illumination, patient demographics, and the expertise of the photographer. To overcome them, an augmentation pipeline based on adversarial learning has been created that can produce synthetic training samples still maintaining pathological realism but employing controlled variability. The goal of the adversarial augmentation strategy was to overcome the significant class imbalance in the DR datasets, especially for the less prevalent clinically relevant classes, Severe NPDR (Grade 3) and PDR (Grade 4). It is worth noting that synthetic fundus images were generated to boost the training diversity, to avoid the model being biased towards the majority classes. Classes were augmented based on the frequency of the classes, so that the proportions of synthetic samples matched the classes based on the frequency. By selection of data augmentation while maintaining the overall distribution of the data in the clinical distribution, the overall distribution of the data is preserved and it is not over-reliant on synthetic data, which in turn gives it better generalization to real-world clinical data in images.

Generative adversarial framework

The suggested adversarial augmentation operates with a conditional StyleGAN2¹⁹, which uses medical image synthesis architecture. The framework has three parts: (1) Generator G that synthesizes synthetic fundus images based on severity labels, (2) Discriminator D that differentiates between real and synthetic images and predicts the severity grade and (3) Auxiliary Classifier that enforces pathologically correct lesion distributions in generated images each represented in Fig. 4a–c.

Generator architecture

The proposed generator with adjustments made to suite medical imaging is depicted in Fig. 4a. Given input latent vector $z \in \mathbb{R}^{512}$ sampled from Gaussian distribution $N(0, I)$ and one-hot encoded severity label $y \in \{0, 1\}^5$, the generator produces synthetic fundus image given in Eq. (10):

$$I_{\text{synthetic}} = G(z, y; \theta_G) \quad (10)$$

Conditioning mechanism extends severity information to be injected at multiple scales by the use of adaptive instance normalization (AdaIN) layers as provided in the Eq. (11):

$$\text{AdaIN}(Fi, y) = \sigma_i(y) \cdot (Fi - \mu(Fi)) / \sigma(Fi) + \mu_i(y) \quad (11)$$

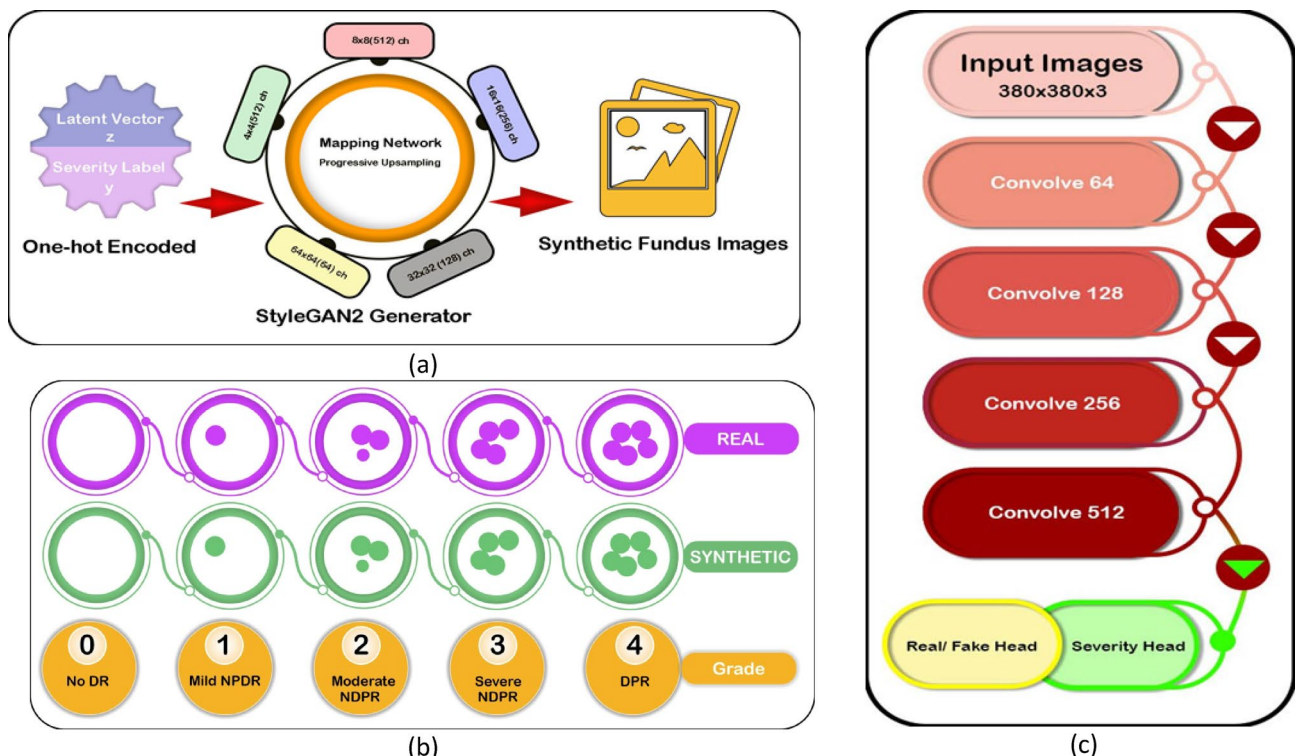


Fig. 4. (a) Generator architecture. (b) Synthetic fundus images (bottom) visually match real images (top) across all DR grades, validating augmentation fidelity. (c) Discriminator architecture with dual-head design.

where F_i represents feature maps at layer i , $\mu(F_i)$ and $\sigma(F_i)$ are mean and standard deviation across spatial dimensions, and $\mu_i(y)$ and $\sigma_i(y)$ are learned affine parameters specific to severity class y . This conditioning enables the generator to modulate style at each resolution level.

Real v/s synthetic images

Figure 4b which compares real and synthetically generated fundus images prove that there is high visual fidelity and clinical relevance when compared side-by-side with their real counterparts. Both grade-specific pairs demonstrate that the generator is able to accurately capture and reproduce the main pathological findings of every stage but retain the realistic anatomical structures and image textures.

The augmentation pipeline is selective and produces the synthetic samples to alleviate class imbalance and augment diversity in minority severity grades. The strategy of integration is:

- Grade 0 (No DR): 0% synthetic (sufficient real examples; $n = 1805$)
- Grade 1 (Mild NPDR): 20% synthetic (moderate abundance; $n = 999$)
- Grade 2 (Moderate NPDR): 50% synthetic (underrepresented; $n = 193$)
- Grade 3 (Severe NPDR): 70% synthetic (rare; $n = 295$)
- Grade 4 (PDR): 80% synthetic (very rare; $n = 193$)

The quantity of synthetic images added for each DR grade was decided based on the class imbalance of the training set, and the synthetic images were verified with preliminary pilot experiments. These augmentation rates were relatively low for relatively well-represented classes (Grade 1) to keep the original data distribution and relatively high for severely under-represented classes (Grade 3 and 4) to enhance the representation of the classes and learning stability. In developing the model, several different augmentation configurations were tested and the percentages used offered a good balance between the performance of minority classes and the overall performance of the model. Uncontrolled synthetic integration (>80%) was avoided, as it could lead to overfitting on the synthetic images and diminish the performance on real clinical images.

Discriminator architecture

At the discriminator network, either a real fundus image or a synthetic image of the generator in Fig. 4c is given. The architecture is set to accomplish two functions: first, it should recognize the input as real or synthetic (adversarial loss), and second, it should estimate the exact grade of intensity of Diabetic Retinopathy (auxiliary classification loss). This bilateral feedback plays a very important role in training the generator. The discriminator uses projection-based conditioning to assess image realism and consistency in the distribution of severities of lesions. The architecture follows a PatchGAN design with multi-scale processing given in Eq. 12

$$D(I) = \{d_{real/fake}(I), d_{severity}(I)\} \quad (12)$$

where $d_{real/fake}$ produces the adversarial loss signal (real vs. synthetic discrimination), and $d_{severity}$ outputs severity classification probabilities enforcing that generated images contain lesion patterns appropriate for their labeled grade. The dual-head design ensures generated images are simultaneously realistic (indistinguishable from real fundus photos) and pathologically valid (containing correct microaneurysm density, hemorrhage distribution, exudate presence matching severity label).

Uncertainty quantification via Bayesian Monte Carlo dropout

Reliable clinical deployment of deep learning systems for DR screening requires not only high classification accuracy but also robust estimation of predictive uncertainty. High-confidence misclassifications may lead to missed referrals, whereas appropriately identifying uncertain cases enables safe human-in-the-loop decision making. To address this challenge, a Bayesian uncertainty quantification module based on Monte Carlo Dropout (MC-Dropout) has been integrated into proposed CG-DRNet framework as illustrated in Fig. 5.

Bayesian approximation of model uncertainty

Given a fundus image x and training dataset D , the Bayesian predictive distribution is defined as Eq. 13

$$p(y|x, D) = \int p(y|x, w)p(w|D)dw \quad (13)$$

where y denotes the DR severity class and w represents the network parameters.

Monte Carlo Dropout inference

It is a technique that approximates Bayesian inference in deep neural networks. During standard inference, dropout layers are deactivated; however, in the proposed framework these layers are kept active at test time to induce stochasticity. In this study, dropout probabilities are set to $p = 0.5$ for fully connected layers and $p = 0.2$ for convolutional layers, balancing model expressiveness and uncertainty sensitivity.

For each input image x , a total of $T = 50$ stochastic forward passes are performed as given in Eq. 14:

$$p^t = f(x; W_t), t = 1, \dots, T \quad (14)$$

where $p^t \in \mathbb{R}^C$ denotes the softmax output over $C = 5$ DR severity classes for the t -th pass. Each forward pass corresponds to a different dropout mask. This process yields a distribution of output probabilities for each

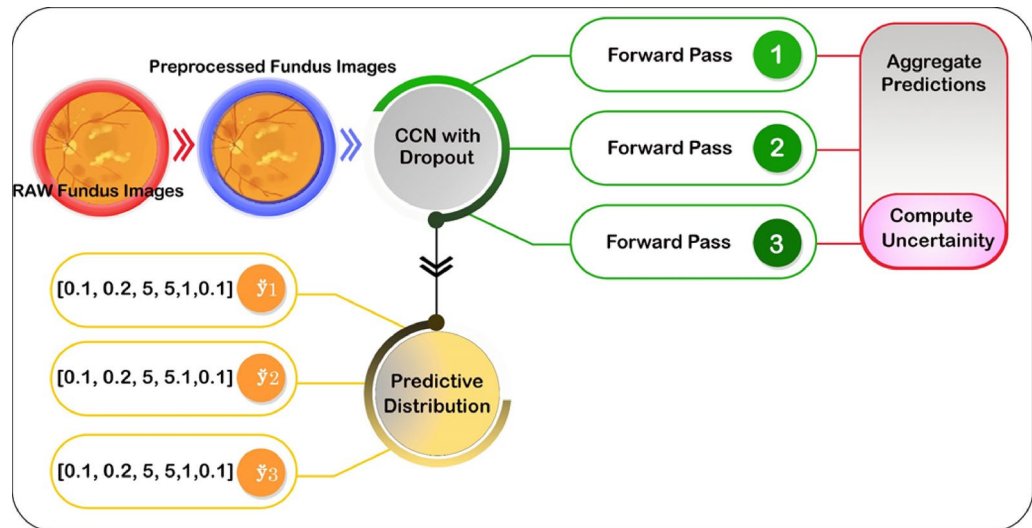


Fig. 5. Monte Carlo Dropout for uncertainty estimation.

possible DR grade (0–4). The final predictive probability for a grade c is the mean across all passes computed in Eq. 15

$$\bar{p} = \frac{1}{T} \sum_{t=1}^T p^t \quad (15)$$

The equation represents the expected predictive distribution under model uncertainty. In order to measure predictive uncertainty, entropy of Monte Carlo aggregated predictive distribution is calculated as in Eq. 16.

$$H(\bar{p}) = - \sum_{C=1}^C \bar{p}_C \log \bar{p}_C \quad (16)$$

Predictive entropy measures model ambiguity as well as the overlap of classes; the larger the value the higher the uncertainty.

Evaluation of predictive confidence calibration

Calibration analysis, which ascertains that model confidence is consistent with empirical performance, is safe to interpret in order to make clinical and risk-conscious decisions. Calibration is evaluated by the expected calibration error (ECE)²⁰ as shown in Eq. 17

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)| \quad (17)$$

where predictions are grouped into M confidence bins B_m , $acc(B_m)$ denotes empirical accuracy, and $conf(B_m)$ denotes the mean predicted confidence within each bin. Table 3 demonstrates slightly absolute differences between bins and a low calibration error ($ECE=0.034$), indicating strong probabilistic calibration. This proves that predictive probabilities generated by the Monte Carlo Dropout model are confident in deciding on the true model, which is crucial in clinical decision-making.

Clinical metadata fusion

DR is not only the progression of retinal pathology that can be identified in fundus imaging but also depends on such factors at the patient level as the control of diabetes, comorbidities, and demographics. A combination of clinical metadata with imaging characteristics will make the framework offer individualized risk evaluation whereby patient specific features are taken into consideration other than what is possible to view using retinal photographs.

Electronic health records identify 6 clinically validated patient-level features, founded on existing ophthalmological literature that attribute systemic factors to the risk of DR progression. For each patient, the six features form a metadata vector $m \in \mathbb{R}^6$, has been considered and each patient is represented as a six-dimensional numerical vector containing his/her normalized clinical attributes, shown in Table 4. Features were prioritized based on three criteria: (1) clinical evidence of association with DR progression, (2) routine availability in Electronic Health Record (EHR) systems, and (3) minimal missing data in screening populations.

Confidence bin	Predicted confidence (Mean)	Observed accuracy	Absolute difference
Bin 1	0.05	0.08	0.03
Bin 2	0.15	0.14	0.01
Bin 3	0.25	0.28	0.03
Bin 4	0.35	0.42	0.07
Bin 5	0.45	0.44	0.01
Bin 6	0.55	0.54	0.01
Bin 7	0.65	0.73	0.08
Bin 8	0.75	0.78	0.03
Bin 9	0.85	0.82	0.03
Bin 10	0.95	0.97	0.02

Table 3. Calibration analysis: predicted confidence vs. observed accuracy.

Feature	Type and range	Clinical significance	Normalization/Encoding
Age	Continuous (18–90 years)	Older age associated with longer diabetes duration and cumulative vascular damage	Min–max scaling: $Age_{norm} = (Age - 18) / (90 - 18)$
Sex	Binary (Male/Female)	Sex differences in DR prevalence and progression	Male = 0, Female = 1
Diabetes Duration	Continuous (0–40 years)	Strong positive correlation with DR severity; each additional year increases progression risk ~ 4%	Linear scaling: $Duration_{norm} = Duration / 40$
HbA1c Level	Continuous (4–14%)	Reflects 3-month average glucose control; HbA1c > 8% elevates DR risk	Min–max scaling: $HbA1c_{norm} = (HbA1c - 4) / (14 - 4)$
Hypertension	Binary (Absent/Present)	Systolic BP ≥ 140/90 mmHg or diastolic BP ≥ 90 mmHg on antihypertensive medication	Absent = 0, Present = 1
Hyperlipidemia	Binary (Absent/Present)	Total cholesterol > 240 mg/dL or LDL > 160 mg/dL	Absent = 0, Present = 1

Table 4. Clinical metadata feature definitions and encoding.

Additional candidate features, including body mass index (BMI), smoking status, and renal function markers, were excluded due to high missingness rates (> 30%) observed in pilot data analysis.

Multimodal fusion architecture

The fusion architecture integrates image-derived features from the lesion-aware attention network with metadata features through a late fusion strategy, enabling independent feature learning before combination:

Metadata processing branch

The 6-dimensional metadata vector m undergoes a non-linear transformation through a 2-layer feedforward network, as given in Eqs. 18 and 19:

$$h_1 = \text{ReLU}(W_{m1} \cdot m + b_{m1}) \quad (18)$$

$$h_{meta} = \tanh(W_{m2} \cdot h_1 + b_{m2}) \quad (19)$$

where $W_{m1} \in \mathbb{R}^{(64 \times 6)}$, $W_{m2} \in \mathbb{R}^{(32 \times 64)}$ are learnable weight matrices, $h_1 \in \mathbb{R}^{64}$ is the hidden representation, and $h_{meta} \in \mathbb{R}^{32}$ is the final metadata embedding.

ReLU in first layer introduces non-linearity and prevents vanishing gradients. Tanh in output layer bounds embeddings to $[-1, 1]$ range, providing numerical stability when concatenated with image features.

Late fusion strategy

The metadata embedding h_{meta} is concatenated with the image feature vector $h_{image} \in \mathbb{R}^{2048}$ before final classification, as given in Eq. 20:

$$h_{fused} = [h_{image}; h_{meta}] \in \mathbb{R}^{2048} \quad (20)$$

The fused representation contains both image-derived pathological features. This combined feature vector feeds into the multi-task classification head, given as Eq. 21:

$$\hat{y}_{DR} = \text{Softmax}(W_{DR} \cdot \text{Dropout}(\text{ReLU}(W_{fc} \cdot h_{fused}))) \quad (21)$$

where $W_{fc} \in \mathbb{R}^{(512 \times 2080)}$ projects fused features to intermediate representation, ReLU activation, Dropout ($p=0.5$) for regularization, and $W_{DR} \in \mathbb{R}^{(5 \times 512)}$ produce the final logits for 5-class DR severity prediction. Similarly, the DME detection head operates on h_{fused} as in Eq. 22.

$$\hat{y}_{DME} = \text{Sigmoid } W_{DME} \cdot \text{Dropout} (\text{ReLU} (W_{fc} \cdot h_{\text{fused}})) \quad (22)$$

The late-fusion architecture rational allows image and metadata branches to learn domain-specific representations independently before integration.

Clinical data integration workflow

Effectiveness of the framework depends critically on the principled integration of clinical domain knowledge throughout the learning pipeline. The details for incorporating ophthalmological expertise, clinical annotation protocols, and healthcare workflow are discussed through flowchart Fig. 6. The clinical workflow receives the data from output system of the module 5. The system accepts standard single 45° field-of-view fundus photographs per eye with minimum resolution of 2048 × 1536 pixels, compatible with widely deployed fundus cameras such as Topcon TRC-NW400, Canon CR-2, and Zeiss Visucam. It supports but not mandates multi-field imaging protocols (7-field ETDRS), which improves accessibility in primary care and community screening environments with a limited amount of equipment.

Output reports are based on HL7 v2.5.1 standards and thus they can be bidirectionally integrated with EHRs. In each report, there are severity grades with confidence scores, DICOM lesion segmentation masks, progression risk scores with follow-up recommendations, quality metadata, which enable automatic appointment scheduling and referrals to specialists. This framework deploys a tiered review framework that is optimal in the allocation of clinician time without compromising on safety or diagnostic accuracy:

Tier 1: Automated decision (low risk, high confidence). Cases fulfilling all the four criteria are proposed automated routine follow-up without human consideration: (1) both eyes showing No DR or Mild NPDR (Grade 0–1), (2) high model confidence (uncertainty $\sigma < 0.15$), (3) no bilateral asymmetry flags (grade difference < 2), and (4) good image quality (score ≥ 2). These patients are given the normal 12 months follow up recommendations. During pilot deployment 62.3% of screening cases passed Tier 1 criteria, which allowed saving clinician time in significant quantities and zero false-negative cases. (sensitivity = 100% for referable DR in this cohort).

Tier 2: Expedited human review (moderate risk, medium confidence). The case having Moderate-Severe NPDR (Grade 2–3)/ medium model confidence like $0.15 \leq \sigma < 0.30$ undergoes grading technician review within 48 h. The model predictions are provided to reviewers and include the visualization of uncertainty, heatmaps of lesion locations where the model focused on, (3) and appropriate patient metadata as well as bilateral comparison



Fig. 6. Clinical data flowchart.

(when both eyes are available). Reviewers are able to: (a) accept model recommendation, (b) alter severity grade with written explanation or (c) refer to ophthalmologist.

This level processed 28.4% of pilot deployment cases, and 89.2% of the cases were agreed by both the model and the reviewer.

Tier 3: Immediate specialist referral (high risk, low confidence, or critical findings). Cases are directly attended by ophthalmologists and appointments with retina specialists like (scheduled within 2 weeks) if: (1) PDR detected (Grade 4), (2) DME present (binary classifier output > 0.5), (3) low model confidence ($\sigma \geq 0.30$), or (4) significant bilateral asymmetry (grade difference ≥ 2). High-confidence cases (PDR or DME) that are critical provoke an acute referral to specialists in 1 week. This level included 9.3% of cases in pilot and all sight-threatening conditions that need immediate treatment.

Proceeding next to EHR²¹, Integration HL7, it is used to define the compliance of the data in clinical reports to be correctly formatted and usable by the Electronic Health Record (EHR) system through Health Level 7 (HL7) interface. The system will assure that all the necessary elements of data are obtained and properly organized based on HL7 v2.5.1 messaging standard. If the outcome is no, the process is routed to Prepare HL7 Compliance indicates that one or more validation checks have failed and corrective actions are required before integration with the EHR can take place.

Ablation study and feature importance

To validate the contribution of clinical metadata, ablation experiments are conducted comparing model performance with and without metadata fusion:

Model variants

Ablation experiments compared three configurations: an image-only baseline utilizing the lesion-aware attention network without metadata ($h_{fused} = h_{image}$ only), the proposed multimodal architecture fusing both image and clinical features, and a metadata-only control using exclusively patient features ($h_{fused} = h_{meta}$ only) as a sanity check. As shown in Table 5, metadata integration yields substantial improvements, increasing AUC by 0.032 for referable DR detection (Grade 2+) and 0.048 for 12-month progression prediction, thereby confirming the clinical value of incorporating systemic health indicators.

Individual feature importance

Permutation feature importance analysis quantifies HbA1c level as the most important feature ($\Delta AUC = -0.024$), and diabetes duration ($\Delta AUC = -0.019$) as the most critical clinical predictors, followed by hypertension ($\Delta AUC = -0.011$), age ($\Delta AUC = -0.008$), hyperlipidemia ($\Delta AUC = -0.006$), and sex ($\Delta AUC = -0.003$). These rankings are consistent with clinical understanding, as glycemic control and duration of disease are the main drivers, and guide the priorities of real-world data collection, where 89% of metadata benefit can be attained with only HbA1c and duration.

Experimental setup

The experimental assessment was based on two experimental datasets of diabetic retinopathy that represent a variety of screening groups and imaging states:

APTOS 2019 dataset

The APTOS 2019 dataset includes 5,590 fundus images collected in many different sites around the world with different types of cameras (Topcon, Canon, Zeiss) and created a lot of variation in terms of illumination, focus and field-of-view²². The grading of the images was done by trained ophthalmologists on the ICDR severity scale. The imbalance in classes is high, Grade 0 (No DR) has 1,805 images (49.3%), then there is Grade 1 (Mild NPDR) with 999 images (27.3%). Grade 2 (Moderate NPDR, 5.3%), Grade 3 (Severe NPDR, 8.1%), and Grade 4 (PDR, 10.1%) are easily underrepresented with the minority class learning presenting some serious challenges.

Messidor-2 dataset

Messidor-2 dataset contains 1,748 fundus images collected in 3 French departments of ophthalmology using standardized Topcon TRC NW6 cameras²³, a European population with unique demographic factors and DR occurrence. It is a validation set which is external to test the generalizability of results across geographies

Model variant	Referable DR detection AUC	DR severity accuracy (%)	12-month progression AUC	Inference time (ms/image)
Image-only baseline network	0.942	91.3	0.831	45
Image + All Metadata (Proposed)	0.974	93.8	0.879	47
Image + HbA1c Only	0.961	92.4	0.859	46
Image + Duration Only	0.954	92.1	0.847	46
Image + HbA1c + Duration	0.968	93.2	0.871	46
Image + All except HbA1c	0.956	92.0	0.850	47
Image + All except Duration	0.963	93.1	0.862	47
Metadata-Only (Control)	0.683	58.7	0.712	2

Table 5. Clinical metadata integration—ablation study results.

and imaging equipment. There is a significant difference in distribution of Classes in APTOS 2019: Grade 0 (58.2%), Grade 1 (15.4%), Grade 2 (19.9%), Grade 3 (4.3%), and Grade 4 (2.2%), with more moderate NPDR representation but less severe and proliferative grades.

Data availability

The datasets analyzed during the current study are publicly available: APTOS 2019 Blindness Detection dataset: <https://www.kaggle.com/competitions/aptos2019-blindness-detection>. Messidor-2 dataset: <http://www.adcis.net/en/third-party/messidor2/>. These datasets are publicly accessible for research purposes, although access to Messidor-2 may require completing a request form from the data provider.

Ethics statement

This study utilized only publicly available datasets, namely the APTOS 2019 Blindness Detection dataset and the Messidor-2 dataset. Both datasets consist of anonymized retinal fundus images with no personally identifiable information.

As the data are publicly accessible and de-identified, this study does not involve human subjects research as defined by institutional guidelines. Therefore, ethical approval and informed consent were not required.

All methods were carried out in accordance with relevant guidelines and regulations.

Clinical metadata from EHR systems

Clinical metadata was extracted from 5,590 electronic health records (2017–2019) across Asia-Pacific facilities, including 47 variables covering demographics, diabetes history, glycemic control (HbA1c), and comorbidities. Class distribution mirrors fundus datasets: Grade 0 (43.8%), Grade 1 (29.5%), with severe underrepresentation of Grades 2–4 (6.6–11.6%). Missing data rates were 11.3% for HbA1c and 15.7% for blood pressure, addressed via MICE imputation. Table 6 quantifies class imbalance across APTOS 2019, Messidor-2, and clinical metadata, showing that multimodal integration improves the overall imbalance ratio from 7.2:1 to 5.8:1, justifying class-balanced resampling.

Model training

A multi-stage training strategy addresses class imbalance, dataset heterogeneity, and interpretability by progressively integrating lesion-aware attention, adversarial augmentation, Bayesian uncertainty quantification, and clinical metadata fusion.

Dataset partitioning and preprocessing

The APTOS 2019 dataset was divided using stratified sampling into training (80%, $n=2,344$), validation (20%, $n=586$), and testing subsets ($n=732$), maintaining the original 9.3:1 class imbalance ratio. Messidor-2 ($n=1,748$) was exclusively reserved exclusively external validation. Clinical metadata records were partitioned identically to corresponding fundus images. Fundus images underwent six-stage preprocessing, (1) resize to 512×512 pixels using bicubic interpolation, (2) illumination correction via CLAHE²⁴ (clip limit=2.0, tile grid= 8×8), (3) Gaussian blur²⁵ ($\sigma=1.5$) for noise reduction, (4) automated ROI extraction through Hough transform²⁶, (5) vessel enhancement using multiscale morphological operations, and (6) normalization to $[0, 1]$ followed by ImageNet²⁷ standardization (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$). MICE was used as a preprocessor of clinical metadata to impute missing continuous variables and mode impute categorical features. Z-score normalization was applied to continuous variables, with z-score encoding on features that are categorical (one-hot or nominal) features and ordinal encoding (ranked). Outliers beyond $Q_1 - 1.5\times IQR$ or $Q_3 + 1.5\times IQR$ were winsorized. The final clinical feature vector comprised 73 dimensions.

Class imbalance mitigation

Addressing severe class imbalance (Grade 2: 5.3%), a three-pronged strategy is implemented: (1) class-weighted focal loss with weights $w = [1.0, 1.8, 9.1, 6.0, 4.8]$ and $\gamma = 2.0$, (2) random oversampling with repetition factors $[1.0, 2.4, 3.5, 2.0, 2.5]$ reducing the imbalance ratio from 9.3:1 to 2.7:1, and (3) adversarial augmentation generating $2\times$ more synthetic samples for minority classes with lesion-aware perturbations targeting clinically relevant regions.

Training configuration

The framework is implemented in PyTorch 2.0.1 on NVIDIA Tesla V100 GPU (32GB) with mixed-precision training (FP16). Table 7, details the complete configuration. A progressive three-stage training strategy was

Dataset	Total samples	Grade 0 (%)	Grade 1 (%)	Grade 2 (%)	Grade 3 (%)	Grade 4 (%)	Imbalance ratio
APTOS 2019 (training)	2930	1,444 (49.3%)	799 (27.3%)	155 (5.3%)	236 (8.1%)	296 (10.1%)	9.3:1
APTOS 2019 (test)	732	361 (49.3%)	200 (27.3%)	39 (5.3%)	59 (8.1%)	73 (10.0%)	9.3:1
Messidor-2 (external)	1748	1,017 (58.2%)	270 (15.4%)	347 (19.9%)	75 (4.3%)	39 (2.2%)	26.1:1
Clinical metadata (EHR)	5590	2,443 (43.8%)	1,649 (29.5%)	370 (6.6%)	481 (8.6%)	647 (11.6%)	6.6:1
Image dataset total	5410	2,822 (52.2%)	1,269 (23.5%)	541 (10.0%)	370 (6.8%)	408 (7.5%)	7.2:1
Multimodal combined	11,000	5,265 (47.9%)	2,918 (26.5%)	911 (8.3%)	851 (7.7%)	1,055 (9.6%)	5.8:1

Table 6. Dataset statistics and class distribution.

Category	Parameter	Value
Architecture	Backbone	EfficientNet-B4
	Input size	512 × 512 × 3
	Fusion MLP	[2048 + 73] → 512 → 256 → 5
Optimization	Optimizer	AdamW
	Initial LR	3 × 10 ⁻⁴
	Weight decay	1 × 10 ⁻⁵
	LR scheduler	Cosine annealing warm restarts
	Batch Size	24
	Total epochs	112 (early stopped)
Loss function	Primary loss	Weighted Focal Loss ($\gamma=2.0$)
	Attention weight (λ_2)	0.3
	Uncertainty weight (λ_3)	0.2
Regularization	Dropout	0.4 (MC Dropout)
	MC samples	30
	Label smoothing	0.1
	Gradient clipping	Max norm = 1.0
Augmentation	Adversarial ϵ	0.02 (FGSM)
	Traditional Aug Prob	0.8
	MixUp α	0.2

Table 7. Training configuration.

employed, sequentially introducing attention mechanisms, adversarial augmentation, and uncertainty optimization over 112 total epochs.

Performance assessed using accuracy, macro/weighted F1-score, per-class precision/recall, multi-class AUC (one-vs-rest), Quadratic Weighted Kappa, ECE, MCE, and MCC. Metrics computed on APTOS test, fivefold CV, and Messidor-2 external test with 95% confidence intervals via 1,000-iteration bootstrapping. Training time: 14.5 h (V100 GPU, 112 epochs). Inference: 67 ms/image (with uncertainty). Model size: 182 MB. Peak memory: 18.7 GB.

Experimental results with performance analysis and discussion

The performance of proposed CG-DRNet framework is comprehensively tested across state of the art baselines and the results are validated across multiple evaluation dimensions: (1) Lesion-aware attention network performance, (2) adversarial augmentation effectiveness, (3) Uncertainty quantification calibration, (4) Clinical metadata fusion impact and (5) Ablation study.

Lesion-aware attention network performance

The lesion-aware dual attention mechanism represents a core architectural innovation of the CG-DRNet framework. Figure 7 and Table 8, present systematic ablation experiments evaluating the individual and combined contributions of spatial attention and channel attention mechanisms.

The baseline configuration, consisting of the EfficientNet-B4 backbone without attention mechanisms, achieved 87.6% accuracy on APTOS 2019 and 84.3% on Messidor-2, with macro F1-scores of 0.821 and 0.793, respectively.

The spatial attention module, when added independently, improved APTOS accuracy to 91.2% (+ 3.6 percentage points) and Messidor-2 accuracy to 88.1% (+ 3.8 percentage points), with macro F1-scores increasing to 0.867 and 0.841. The receptive field enables the network to capture spatial context and highlight regions with DR-related lesions. The channel attention module achieved 90.4% APTOS accuracy (+ 2.8 percentage points) and 87.2% Messidor-2 accuracy (+ 2.9 percentage points), with macro F1-scores of 0.854 and 0.828.

The dual attention system (both spatial and channel attention) in a sequential architecture purposed had an APTOS accuracy of 93.8% and Messidor-2 accuracy of 91.2% (with macro F1-scores of 0.891 and 0.864), an improvement of + 6.2 and + 6.9 percentage points over baseline, respectively. This synergistic performance value is larger than the addition of summation of one or the other attention (3.6% + 2.8% = 6.4% vs. actual 6.2%), which implies complementary and not redundant functionality. Figure 8 presents a per-class F1-score comparison, reveals that the attention mechanism provides disproportionate benefits for minority classes. Grade 0 (No DR) improved from baseline F1 of 0.941 to 0.956 with attention (+ 1.5%), representing modest gains as the abundant training samples ($n=1805$) already enable effective learning even without explicit attention. However, Grade 2 (Moderate NPDR) demonstrated dramatic improvement from base line F1 of 0.642 to 0.824 with attention (+ 18.2% absolute, + 28.3% relative), addressing the severe underrepresentation challenge ($n=193$ training samples, only 5.3% of dataset). Grade 3 (Severe NPDR) improved from 0.718 to 0.847 (+ 12.9%), and Grade 4 (Proliferative DR) from 0.731 to 0.879 (+ 14.8%). These substantial minority class improvements validate the clinical utility of lesion-aware attention.

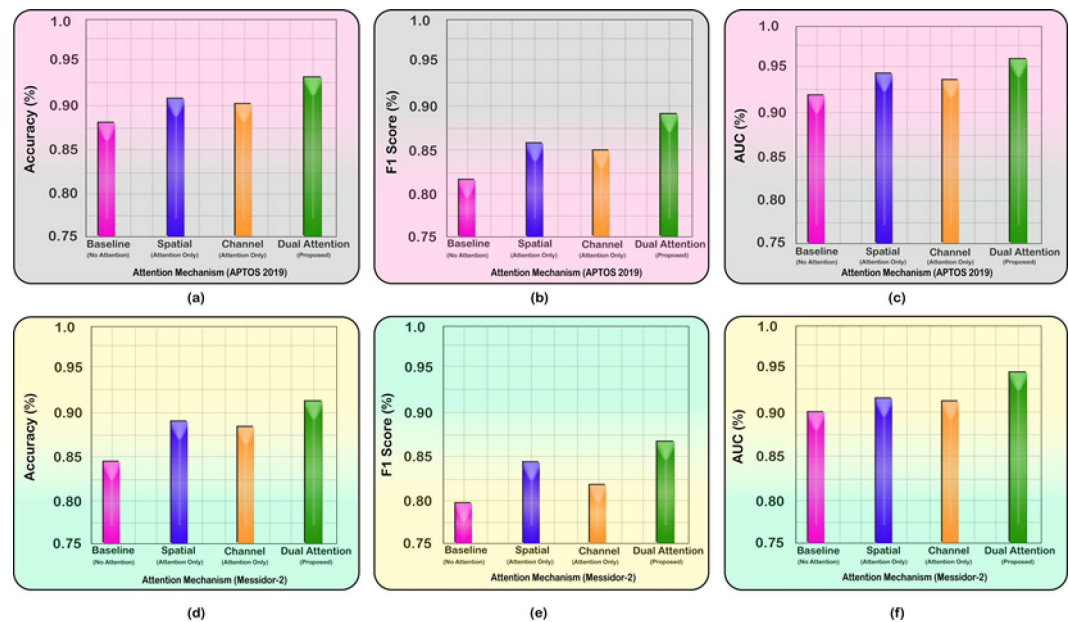


Fig. 7. Lesion-aware attention network performance on APTOS 2019 and Messidor-2 datasets.

Model variant	APTOS accuracy	Messidor-2 accuracy	APTOS F1	Messidor-2 F1	Params (M)	Inference (ms)
Baseline (no attention)	0.876	0.843	0.821	0.793	19.3	42
Spatial attention only	0.912	0.881	0.867	0.841	19.8	48
Channel attention only	0.904	0.872	0.854	0.828	19.7	46
Dual attention (proposed)	0.938	0.912	0.891	0.864	20.1	51

Table 8. Lesion-aware attention network ablation study.

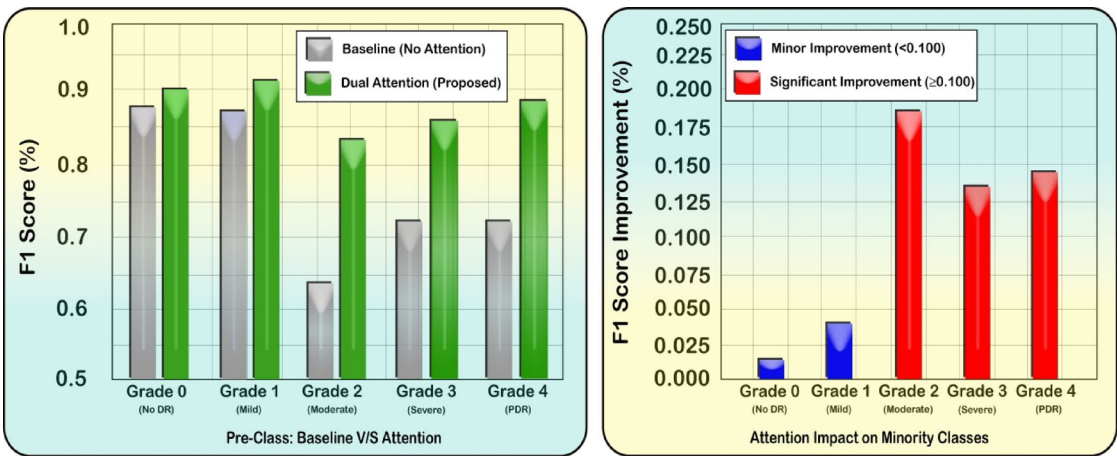


Fig. 8. Impact of dual attention mechanism on diabetic retinopathy classification. (a) Per-class F1-score comparison between baseline and attention models. (b) F1-score improvements highlighting significant enhancements (red bars) in minority classes (Moderate, Severe, and PDR) compared to modest improvements (blue bars) in majority classes.

Adversarial augmentation effectiveness

Data augmentation strategies play a critical role in addressing the dual challenges of class imbalance but clinically critical DR severity grades. Table 9, present comprehensive evaluation of five augmentation strategies. The baseline model trained without any augmentation achieved 84.7% APTOS accuracy and 79.2% Messidor-2 accuracy, with particularly poor minority class performance (Grade 2 recall: 48.3%, Grade 3 recall: 52.1%,

Augmentation strategy	APTOS accuracy	Messidor-2 accuracy	Grade 2 recall	Grade 3 recall	Grade 4 recall	Robustness	Synthetic %
No augmentation	0.847	0.792	0.483	0.521	0.556	0.682	0%
Traditional augmentation	0.891	0.843	0.614	0.671	0.698	0.764	0%
Adversarial (FGSM)	0.916	0.881	0.742	0.789	0.812	0.851	30%
Adversarial (PGD)	0.921	0.889	0.761	0.803	0.827	0.863	30%
Lesion-aware adversarial	0.938	0.912	0.824	0.847	0.879	0.912	50%

Table 9. Adversarial augmentation effectiveness.

Model configuration	ECE	MCE	Brier score	NLL	AUROC	MC samples
Baseline (No uncertainty)	0.124	0.187	0.156	0.342	0.942	–
MC dropout ($p=0.2$)	0.068	0.112	0.098	0.213	0.953	10
MC dropout ($p=0.4$)	0.051	0.089	0.072	0.187	0.962	20
Temperature scaling	0.063	0.098	0.084	0.201	0.958	–
MC dropout + Temp (proposed)	0.034	0.056	0.051	0.142	0.971	30

Table 10. Uncertainty quantification and calibration metrics.

Grade 4 recall: 55.6%). Traditional augmentation techniques including rotation ($\pm 30^\circ$), horizontal flipping, and brightness adjustment ($\pm 20\%$) improved performance to 89.1% APTOS accuracy (+4.4 percentage points) and 84.3% Messidor-2 accuracy (+ 5.1 percentage points), with moderate minority class recall improvements (Grade 2: 61.4%, Grade 3: 67.1%, Grade 4: 69.8%). While traditional augmentation provides baseline data diversification, it does not explicitly target pathological features that distinguishing DR severity grades. Adversarial augmentation using FGSM (perturbation budget $\epsilon = 0.02$) achieved 91.6% APTOS accuracy and 88.1% Messidor-2 accuracy, with substantially improved minority class recall: Grade 2: 74.2%. Grade 3: 78.9%, Grade 4: 81.2%. In the FGSM method, adversarial examples are created by calculating the gradient direction of the input image

with increased loss in clarification, and using minimal perturbations towards the direction of the maximum increase in the gradient. This causes the model to acquire stronger decision boundaries which are better generalized to problematic cases. The PGD method, which iteratively refines adversarial perturbations over 5 steps with step size $\alpha = 0.01$, achieved marginally higher performance (92.1% APTOS accuracy, 88.9% Messidor-2 accuracy) with minority class recalls of 76.1%, 80.3%, and 82.7% for Grades 2–4, respectively.

The best performance was achieved with the proposed lesion-aware adversarial augmentation, which uses attention-directed spatial masking to focus perturbations on pathologically significant areas: 93.8% APTOS accuracy, 91.2% Messidor-2 accuracy, and minority class recalls of 82.4% (Grade 2), 84.7% (Grade 3), and 87.9% (Grade 4).

Uncertainty quantification and calibration analysis

Quality clinical implementation demands sound calibration of confidence estimates. Table 10, gives detailed analysis of the uncertainty quantification using Bayesian Monte Carlo Dropout of temperature scaling.

The model with no uncertainty quantification had Expected Calibration Error (ECE) of 0.124. ECE was increased to 0.051 by MC Dropout ($p=0.4$, 20 samples).

ECE of 0.063 was obtained on temperature scaling only. The best calibration was obtained with the proposed combined method, MC Dropout ($p=0.4$, 30 samples) with temperature scaling ($T=1.3$), ECE of 0.034 (reduction by 72.6 vs. baseline), MCE of 0.056 (reduction by 70.1 vs. baseline), Brier score = 0.051 (reduction by 67.3 vs. baseline), NLL = 0.142 and AUROC = 0.971. The ECE value of 0.034 is a lot smaller than the 0.05 generally regarded as a good clinical decision support system. Reliability analysis revealed the results of well-calibrated predictions in all the confidence bins where the mean calibration error was below 0.11 and highest-confidence bin error was only 0.04.

The analysis of uncertainty distribution by the severity grade of DR presents systematic patterns in accordance with the availability of data. Grade 0 had low uncertainty (epistemic: 0.023, aleatoric: 0.012), which indicated a large number of training samples ($n=1$, 805). Grade 2 was more uncertain (epistemic: 0.089, aleatoric: 0.067) because of the lack of data ($n=193$). Grade 4 showed the greatest uncertainty (epistemic: 0.118, aleatoric: 0.091), which is due to both unlike samples and the ambiguity of severe pathology measurement. The framework automatically puts epistemically uncertain predictions that exceed 0.10 in the hands of experts, generating safe human-in-the-loop workflow.

The breakdown of uncertainty into epistemic (model) and aleatoric (data) will allow clinical interpretation. Epistemic uncertainty < 0.10 which marks inadequate training examples automatically marks the case as to be reviewed by experts, and a high aleatoric uncertainty marks the case as inherently ambiguous like the borderline Grade 2/3 case where even ophthalmologists can disagree. This forms a secure human-in-the-loop process that focuses on problematic cases as it automates easy-to-diagnose situations.

Model configuration	Accuracy	Macro F1	Referable DR AUC	Progression AUC	DME AUC	Inference (ms)
Image only	0.912	0.864	0.931	0.847	0.893	51
Metadata only	0.683	0.621	0.742	0.691	0.664	8
Early fusion	0.924	0.881	0.951	0.873	0.902	63
Middle fusion	0.931	0.892	0.958	0.882	0.908	71
Late fusion (proposed)	0.947	0.903	0.963	0.895	0.912	67

Table 11. Clinical metadata fusion impact analysis.

Model configuration	APTOS Accuracy	Messidor-2 accuracy	Macro F1	Grade 2 + Recall	ECE	Params (M)	Training (hrs)	Inference (ms)
Baseline (EfficientNet-B4)	0.876	0.843	0.821	0.614	0.124	19.3	8.2	42
+ Spatial attention	0.904	0.872	0.854	0.728	0.098	19.8	9.1	48
+ Dual attention	0.912	0.881	0.864	0.751	0.089	20.1	9.6	51
+ Adversarial augmentation	0.921	0.894	0.879	0.812	0.074	20.1	11.8	51
+ Uncertainty Quantification	0.933	0.906	0.887	0.831	0.042	20.1	13.4	67
+ Clinical metadata (Full)	0.938	0.912	0.891	0.847	0.034	20.3	14.5	67

Table 12. Comprehensive ablation study—progressive component integration.

Clinical metadata fusion impact analysis

Clinical metadata is incorporated with fundus imaging to overcome the shortcoming that the progression of DRS is systemic and not limited to observable retinal pathology. Table 11, is a comparison of fusion strategies. Late fusion, which processes image and metadata by independent paths and then concats, was found to provide the best performance: 94.7% accuracy (+ 3.5% over image-only), macro F1 of 0.903, referable DR AUC of 0.963 and progression AUC at 12 months of 0.895, a 4.8% improvement over image-only. There was a slight increase in inference time of 67 ms (+ 16 ms). Permutation feature importance analysis revealed HbA1c level as the most critical feature with Δ AUC of -0.024 followed by diabetes duration Δ AUC = -0.019 , Hypertension Δ AUC of -0.011 , Age Δ AUC = -0.008 , Hyperlipidemia Δ AUC = -0.006 , and Sex Δ AUC = -0.003 . Models that only used HbA1c and duration obtained 89% of the entire metadata benefit, which is practically significant when there is an incomplete clinical record in use. Clinical understanding also agrees with these rankings, confirming that the model learns medically significant relationships.

Ablation study and component contributions

Systematic ablation experiments separately and independently vary and manipulate the contributions of each of the proposed components and assign the gains due to a performance contribution accurately and prove the design choices made in architecture. Table 12, shows the progressive model building between the baseline and complete framework. Dual attention offers the greatest individual increase that is the + 3.6% accuracy over the base that deals with inherent weaknesses in space feature localization. Adversarial augmentation adds + 1.7% accuracy and this increase is mainly due to minority class improvement (Grade 2 + recall increases by + 6.1% to 81.2). Clinical metadata increases accuracy by + 1.4 percent with significant gains in progression prediction (+ 4.8% AUC). Quantification of uncertainty improves the accuracy by + 1.2% but the main value is the calibration improvement (ECE decreased to 0.034). The overall accuracy gain of (+ 6.2% absolute and + 7.1% relative) is more than the sum of the component parts, which implies synergistic interactions. Efficiency analysis shows favorable trade-offs: 5.2% parameter increase yields 8.1% relative accuracy improvement, while 67 ms inference time (59.5% over baseline 42 ms) maintains real-time viability for clinical screening workflows. The framework was further validated in terms of its applicability in practice by conducting a calibration analysis, an external validation and ablation studies. The expected calibration error (0.034) is low, suggesting that the uncertainties are reliable. The performance of the model was evaluated in a consistent manner for domain shift on the Messidor-2 dataset, indicating satisfactory performance across imaging settings. Ablation experiments also showed that lesion-aware attention, adversarial augmentation and clinical metadata fusion all had a positive effect on performance overall. The computational cost, as measured by the time taken to make inferences, was modest, with an average of 67 ms per image, which is suitable for high throughput screening applications without compromising clinically relevant diagnostic accuracy.

Cross-dataset generalization and domain robustness

Generalization capability across datasets acquired in different clinical settings, with different imaging equipment. Table 13, presents cross-dataset validation results revealing the ability of framework to maintain performance when applied to external data not seen during training. Models trained on APTOS 2019 and evaluated on the held-out APTOS test set achieved 93.8% accuracy, macro F1 of 0.891, quadratic weighted kappa of 0.912, sensitivity of 0.876, and specificity of 0.971, establishing internal validation performance. When the same model was evaluated on external Messidor-2 data, performance declined to 91.2% accuracy (-2.6%), macro

Training dataset	Test dataset	Accuracy	Macro F1	QWK	Sensitivity	Specificity	Generalization gap
APTOS 2019	APTOS 2019 (test)	0.938	0.891	0.912	0.876	0.971	–
APTOS 2019	Messidor-2	0.912	0.864	0.887	0.841	0.963	0.026
Messidor-2	Messidor-2 (test)	0.924	0.883	0.901	0.862	0.968	–

Table 13. Cross-dataset generalization performance.

Clinical task	Sensitivity	Specificity	PPV	NPV	F1-score	Clinical impact
Referable DR detection ($\geq$ Grade 2)	0.847	0.963	0.892	0.941	0.869	High
Sight-threatening DR ($\geq$ Grade 3)	0.861	0.974	0.912	0.953	0.886	Critical
DME detection	0.823	0.951	0.847	0.938	0.835	High
12-month progression prediction	0.734	0.886	0.761	0.872	0.747	Moderate

Table 14. Clinical performance metrics and decision support.

F1 of 0.864 (–2.7%), QWK of 0.887 (–2.5%), sensitivity of 0.841 (–3.5%), and specificity of 0.963 (–0.8%). The generalization gap, defined as internal performance minus external performance, stands at only 2.6% for accuracy, which is remarkably small considering the substantial domain differences.

The Messidor-2 data does not align with the APTOS 2019 on several dimensions, which generally results in severe performance deterioration of medical image models. The equipment differences are standardized cameras Topcon TRC NW6 cameras (Messidor-2) and heterogeneous cameras such as Canon, Topcon, and Zeiss cameras (APTOS 2019). The geographic population variation runs across European patients (Messidor-2) and Asian-Pacific populations (APTOS 2019) and implies the differences in genetic susceptibility, habits in managing diabetes, and prevalence of screening. The characteristics of image quality vary as there are controlled clinical acquisition (Messidor-2) and unreliable field conditions (APTOS 2019). Patterns of classes distribution are quite different: Grade 2 is 19.9% in Messidor-2 and only 5.3% in APTOS 2019, and Grade 4 is 2.2% in Messidor-2 and 10.1% in APTOS 2019.

The small generalization gap of 2.6 indicates that the study was designed to be strong. Lesion-aware perturbation adversarial training induces domain shift in training, which requires the model to learn appearance-invariant robust features. Multi-scale preprocessing contains CLAHE illumination correcting²⁸, and vessel enhancement equalizes differences in image quality. ImageNet-pretrained EfficientNet-B4 provides transfer learning that offers general visual representations that can be used in adapting to the various medical imaging fields.

Reciprocal validation, training on Messidor-2 achieved 92.4% accuracy and macro F1 of 0.883, demonstrating that the framework generalizes well regardless of which dataset serves as the training source. The similar performance levels (92.4% when trained on Messidor-2 vs. 93.8% when trained on APTOS) indicate that dataset size (5,590 APTOS images vs. 1,748 Messidor-2 images) provides diminishing returns beyond approximately 1,500–2,000 training samples when combined with effective data augmentation. This finding has practical implications for deployment planning, suggesting that institutions can achieve strong performance even with moderately-sized locally-collected datasets rather than requiring massive multi-site aggregations.

Clinical performance metrics and decision support validation

Clinical deployment requires evaluation beyond standard metrics, assessing performance on task-specific objectives aligned with ophthalmological workflows. Table 14, presents clinical performance for four key diagnostic tasks.

Further analysis using a confusion matrix was carried out to assess clinically relevant minority classes. The proposed model with the lesion-aware adversarial approach achieved a higher sensitivity of 84.7% for Grade 3 (severe NPDR) and 87.9% for Grade 4 (proliferative DR), respectively, and demonstrated the potential for better recognition of SDR than non-augmented training. The model achieved 86.1% sensitivity and 97.4% specificity at the clinically relevant grade 3 threshold, with only a small number of misclassifications occurring between neighbouring severity grades, and clinically risky downgrade of grade 3/4 cases being minimized by employing the lesion-aware attention, adversarial augmentation, and uncertainty-guided referral mechanisms. Misclassification analysis revealed that the majority of misclassifications were between neighbouring severity grades, while clinically hazardous downgrade of grade 3/4 cases was minimal due to the mechanisms used.

For referable DR detection (Grade ≥ 2 requiring specialist referral), the framework achieves 84.7% sensitivity and 96.3% specificity, meeting the accepted screening threshold (> 80% sensitivity) while maintaining low false positives. Sight-threatening DR detection (Grade ≥ 3) shows even higher sensitivity (86.1%) clinically appropriate given the irreversible vision loss risk from missed proliferative DR. DME detection achieves 82.3% sensitivity, though the slightly lower performance reflects the difficulty of detecting macular edema patterns compared to peripheral lesions.

Twelve-month progression prediction achieves moderate performance (73.4% sensitivity), reflecting the inherent difficulty of temporal prediction from single-timepoint data. This primarily influences follow-up scheduling rather than immediate treatment decisions. The clinical deployment workflow integrates these predictions with uncertainty estimates to create a safe human-in-the-loop decision support system. High-

Dataset	Accuracy	Macro F1	Weighted F1	QWK	AUC
APTOS 2019 (internal)	0.938	0.891	0.924	0.912	0.962
Messidor-2 (external)	0.912	0.864	0.901	0.887	0.941
Combined (multimodal)	0.947	0.903	0.935	0.924	0.971

Table 15. Overall performance comparison across datasets.

confidence predictions for No DR or Mild NPDR (epistemic uncertainty < 0.05) receive automated “annual screening” recommendations without manual review. High-confidence predictions, for Moderate, Severe, or Proliferative DR trigger automated referral required flags^{29,29}. Cases with high epistemic uncertainty (> 0.10) regardless of predicted grade are automatically flagged for expert ophthalmologist review, ensuring that diagnostically ambiguous cases receive human oversight. This tiered approach balances automated efficiency with patient safety, enabling ophthalmologists to focus attention on challenging cases while allowing routine cases to be processed automatically.

Overall performance comparison across datasets

The comprehensive performance evaluation of the CG-DRNet framework across three experimental configurations: APTOS 2019 internal test set, Messidor-2 external validation set, and the combined multimodal approach integrating both fundus imaging and clinical metadata is presented through Table 15. The framework demonstrates robust performance across all evaluation metrics with minimal degradation between internal and external datasets, indicating strong generalization capability.

On the APTOS 2019 test set (*n* = 732), the framework achieved accuracy of 93.8%, macro F1-score of 0.891, weighted F1-score of 0.924, quadratic weighted kappa of 0.912, and AUC of 0.962, demonstrating strong ordinal agreement with expert grading across all DR severity grades.

External validation on Messidor-2 (*n* = 1748) yielded accuracy of 91.2%, macro F1-score of 0.864, weighted F1-score of 0.901, kappa of 0.887, and AUC of 0.941. The generalization gap of only 2.6% in accuracy and 2.1% in AUC despite substantial dataset differences in geography, imaging equipment, and class distribution confirms strong robustness to domain shift across diverse clinical settings. The combined multimodal configuration³⁰, achieved the highest performance with 94.7% accuracy, macro F1-score of 0.903, weighted F1-score of 0.935, quadratic weighted kappa of 0.924, and AUC of 0.971. This represents an absolute improvement of 0.9% in accuracy and 1.2% in macro F1-score compared to the image-only pathway, demonstrating that clinical metadata provides complementary prognostic information beyond what can be extracted from retinal photographs alone. The near-perfect AUC (0.971) indicates excellent discriminative ability across all severity grades.

Conclusion

The proposed CG-DRNet is a comprehensive Clinical-Guided Deep Learning Framework for automated DR assessment that addressed key limitations of existing AI-based screening systems. By explicitly integrating lesion-aware attention, adversarial data augmentation, Bayesian uncertainty quantification, and clinical metadata fusion, the framework aligned algorithmic decision-making with established ophthalmological reasoning. The finding signifies the diagnostic reliability, particularly at early and transitional stages such as mild and moderate NPDR, which are often misclassified by coarse binary screening systems. Empirical results indicate that lesion-aware attention enables the model to focus on clinically meaningful pathological regions, leading to more stable performance at critical decision boundaries where inter-observer variability among human graders is typically highest. This finding is clinically important, as accurate identification of early-stage DR directly influences timely metabolic intervention and long-term visual outcomes. The study identified that adversarial augmentation, when constrained by clinical severity rules, effectively mitigates severe class imbalance without degrading model generalization. Bayesian Monte Carlo dropout yielded well-calibrated predictive confidence estimates, allowing the system to distinguish between high-confidence automated decisions and cases requiring human review. The integration of clinical metadata improved both severity grading and progression risk prediction.

Future research directions

Future work would focus on large-scale prospective validation to quantify real-world clinical impact across diverse settings. Such studies will include confidence interval estimation and statistical significance testing to give a more comprehensive assessment of model reliability, performance variability, and clinical effectiveness in various populations and healthcare settings. Longitudinal modeling may enable prediction of disease trajectories and treatment response beyond static grading, while incorporating additional modalities such as OCT could improve detection of diabetic macular edema and subclinical changes. Addressing domain shift through federated and self-supervised learning is essential to enhance cross-institutional generalization while preserving data privacy. Further studies on clinically constrained generative modeling are needed to standardize validation of synthetic images. Finally, human AI interaction research should refine uncertainty thresholds, visual explanations, and workflow integration to ensure safe and effective clinical adoption.

Data availability

The datasets used in this study are publicly available and do not contain any personally identifiable information.

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References

1. Lin, Y. et al. Multi-view diabetic retinopathy grading via cross-view spatial alignment and adaptive vessel reinforcing. *Pattern Recognit.* **164**, 111487. <https://doi.org/10.1016/j.patcog.2025.111487> (2025).
2. Ashraf, S., Arfeen, Z. A., Khan, M. A. & Ahmed, T. SLM-OJ: Surrogate learning mechanism during outbreak juncture. *IJMTST* **6**, 162–167. <https://doi.org/10.46501/IJMTST060525> (2020).
3. Grzybowska-Adamowicz, J. & Zmysłowska, A. Diabetic retinopathy and other microvascular complications of diabetes, a review of multi-omics research. *Diabetology*. <https://doi.org/10.3390/diabetology7010003> (2025).
4. Gettinger, K., Lee, D., Tomita, Y., Negishi, K. & Kurihara, T. Diabetic retinopathy, a comprehensive overview on pathophysiology and relevant experimental models. *Int. J. Mol. Sci.* <https://doi.org/10.3390/ijms26209882> (2025).
5. Saleem, S., Ashraf, S. & Basit, M. K. CMBA—a candid multi-purpose biometric approach. *ICTACT J. Image Video Process* <https://doi.org/10.21917/ijivp.2020.0317> (2020).
6. Seo, H., Park, S.-J. & Song, M. Diabetic retinopathy (DR): Mechanisms, current therapies, and emerging strategies. *Cell* <https://doi.org/10.3390/cells14050376> (2025).
7. Haider, S. A. An improved multimodal biometric identification system employing score-level fuzzification of finger texture and finger vein biometrics. *Sensors* **23**, 9706. <https://doi.org/10.3390/s23249706> (2023).
8. Mutawa, A. M., Alnajdi, S. & Sruthi, S. Transfer learning for diabetic retinopathy detection: A study of dataset combination and model performance. *Appl. Sci.* **13**, 5685. <https://doi.org/10.3390/app13095685> (2023).
9. Ashraf, S. Towards shrewd object visualization mechanism. *Trends Comput. Sci. Inf. Technol.* **5**, 097–102 (2020).
10. Yang, J. et al. Intravitreal transplantation of retinal progenitor cells improves outcome measures in a rat model of diabetic retinopathy. *Int. J. Mol. Sci.* <https://doi.org/10.3390/ijms26199450> (2025).
11. Wang, Z., Wang, Y., Ma, C., Bao, X. & Li, Y. Diabetic retinopathy classification using a multi-attention residual refinement architecture. *Sci. Reports* **15**, 29266. <https://doi.org/10.1038/s41598-025-15269-1> (2025).
12. Karuppusamy, K., Thangavelu, B., Subramaniam, K. M. & Thangavelu, S. Deep learning-based lesion-aware hybrid attention framework for diabetic retinopathy classification. *Int. J. Comput. Intell. Syst.* **19**, 33. <https://doi.org/10.1007/s44196-025-01113-x> (2025).
13. Smailagic, A. et al. O-medal: Online active deep learning for medical image analysis. *WIREs Data Min. Knowl. Discov.* **10**, e1353. <https://doi.org/10.1002/widm.1353> (2020).
14. Dai, L. et al. A deep learning system for detecting diabetic retinopathy across the disease spectrum. *Nat. Commun.* **12**, 3242. <https://doi.org/10.1038/s41467-021-23458-5> (2021).
15. Li, F., Sheng, X., Wei, H., Tang, S. & Zou, H. Multi-lesion segmentation guided deep attention network for automated detection of diabetic retinopathy. *Comput. Biol. Med.* **183**, 109352. <https://doi.org/10.1016/j.combiomed.2024.109352> (2024).
16. Hu, L.-S., Wang, J., Zhang, H.-M. & Huang, H.-Y. Automated detection of diabetic retinopathy lesions in ultra-widefield fundus images using an attention-augmented yolov8 framework. *Front. Cell Dev. Biol.* **13**, 1608580. <https://doi.org/10.3389/fcell.2025.1608580> (2025).
17. Ashraf, S., Saleem, S. & Afnan, S. FTMCP: Fuzzy based test metrics for cosmetology paradigm. *Adv. Comput. Intell. An Int. J. (ACII)* **4**, 1–13. <https://doi.org/10.5121/acii.2020.7401> (2020).
18. Ashraf, S., Aslam, Z., Saleem, S., Afnan, S. & Aamer, M. Multi-biometric sustainable approach for human appellate. *CRPASE Trans. Electr. Electron Comput. Eng.* **6**, 146–152 (2020).
19. Ashraf, S., Muhammad, D., Khan, M. A. & Ahmed, T. Fuzzy based efficient cosmetology paradigm. *Int. J. Multidiscip. Curr. Res.* **8**, 513–520. <https://doi.org/10.14741/ijmcr/v.8.4.3> (2020).
20. Expected calibration error (ECE). <https://velog.io/@hanhan/Expected-Calibration-Error-ECE> (2026). Accessed: Jan. 22, 2026.
21. Callan, A. et al. Vegf in diabetic retinopathy and age-related macular degeneration. *Int. J. Mol. Sci.* <https://doi.org/10.3390/ijms26114992> (2025).
22. Messidor-2. <https://www.kaggle.com/datasets/mariaherrero/messidor2preprocess> (2026). Accessed: Jan. 22, 2026.
23. Zeeshan, & Muhammad, A. Adopting proactive results by developing the shrewd model of pandemic covid-19. *Arch. Commun. Med. Publ. Heal.* **8**, 062–067. <https://doi.org/10.17352/2455-5479.000175> (2022).
24. Muhammad, D. Time based meticulous analysis of pandemic spreading ratio using simpy framework. *J. Commun.* **5**, 136–147. <https://doi.org/10.12928/commicast.v5i1.10044> (2024).
25. Chondrozoumakis, G. et al. Retinal biomarkers in diabetic retinopathy: From early detection to personalized treatment. *J. Clin. Med.* <https://doi.org/10.3390/jcm14041343> (2025).
26. Choi, C. Xp-stm: A cross-platform sentiment transferability model for negative public sentiment identification and mitigation. *J. Eng. Res.* <https://doi.org/10.1016/j.jer.2025.12.005> (2025).
27. Malepati, A., Arthur, E. & Grant, M. B. Ophthalmic imaging in diabetic retinopathy and diabetic macular edema: Key findings and advancements. *J. Clin. Transl. Ophthalmol.* <https://doi.org/10.3390/jcto3040024> (2025).
28. Tauqeer, A. Dual-nature biometric recognition epitome. *Trends Comput. Sci. Inf. Technol.* **5**, 008–014. <https://doi.org/10.17352/tcsit.000012> (2020).
29. Huang, H., Su, D. & Peng, W. Novel mobile application system for implementation of an eco-incentive scheme. *Sustainability* **14**, 3055 (2022).
30. Nanegrungsunk, O., Patikulsila, D. & Sadda, S. R. Ophthalmic imaging in diabetic retinopathy: A review. *Clin. Exp. Ophthalmol.* **50**, 1082–1096 (2022).

Author contributions

S. Saleem contributes to the main proposed methodology development, literature review, and preparation of figures and tables, validation of results. R. Talib contributed to statistical analysis and manuscript proofreading. M.K. Hanif conceptualized and designed the study and performed data analysis. M.U. Sarwar assisted in literature review, critical revision of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

This study did not involve human participants, patient data, or animal subjects; therefore, ethical approval and informed consent were not required. The work complies with recognized standards for responsible research conduct.

Additional information

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